Navigating the Privacy–Robustness–Performance Trilemma in Federated Learning

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[2–4 sentences summarizing the paper: motivation (the trilemma), what we did (8-configuration empirical study on MNIST/CIFAR-10 with Flower+Opacus+BLADES), key finding (compounding behavior, direction of magnitude), and main take-away for practitioners.]

Contents

List of abbreviations

FL Federated Learning

IID independent and identically distributed

DP-SGD Differentially Private Stochastic Gradient Descent

SGD Stochastic Gradient Descend

DP Differential Privacy

1 Introduction

We present a systematic empirical study of the Privacy–Robustness–Performance trilemma in Federated Learning (FL), examining how simultaneously deployed defenses interact and compound across all three axes.

[Summarize results and contributions.]

1.1 Background and Motivation

In the past ten years, FL has emerged as a framework for achieving distributed training without the need for data centralization. This is of particular interest when processing sensitive data, such as in healthcare or mobile applications. The local training allows to leave the data with the clients but opens new attack surfaces for model poisoning during training through the transmission of manipulated updates. Also, the transmitted gradients can still leak information about the used training data set. Additionally, gradient transmission at each round causes significant communication cost, especially for complex models.

In an effort to fortify the resilience of models and safeguard client privacy, numerous methodologies have been implemented in recent years. Robustness-enhancing methods are designed to detect malicious updates by assuming a significant deviation from the honest majority. Privacy-preserving algorithms add a mask or calibrated noise to the local updates before transmitting them to the central server.

1.2 Problem Statement

Ensuring model robustness and client privacy while balancing communication cost and model accuracy has been identified as a significant challenge. Adding privacy-enhancing noise to individual updates reduces a malicious server's ability to extract information about local datasets, while simultaneously reducing model accuracy. Discarding unusual updates improves robustness against Byzantine adversaries but risks disregarding legitimate updates, particularly in settings where data is not independent and identically distributed (IID). Compression techniques that reduce transmission load limit the information transmitted to the server and thus reduce model accuracy.

In 2023, Allouah et al. [3] proved Theorem 1, stating that the combination of such methods yields a lower bound for the expected training error that exceeds the sum of the individual cost for robustness and privacy by a multiplicative term.

Theorem 1. *Consider distributed mean estimation in $\mathbb{R}^d$ with n users, of which an η-fraction may behave maliciously. Any algorithm, producing an estimate $\hat{\mu}$ of the true mean $\mu \in \mathbb{R}^d$ of inputs, and achieving (ε, δ)-local DP and robustness to η-corruption must incur an estimated error*

$$\mathbb{E} \parallel \hat{\mu} - \mu \parallel_2^2 = \tilde{\Omega}\left(\frac{\eta}{\varepsilon^2} + \frac{d}{\varepsilon^2 n} + \eta G^2\right). \tag{1}$$

A position paper by the same group [4] highlights a gap between the theoretical lower bound and current implementations.

1.3 Research Questions

The project is structured around three research questions:

1. **Isolation:** What is the individual cost of each defense mechanism (DP, Byzantine-robust aggregation, gradient compression) in terms of accuracy, communication overhead, and attack success rate, and do our results reproduce the published baselines?

2. **Combination:** How do pairwise and full combinations of the three mechanisms interact? Do the costs compound additively, sub-additively, or super-additively?

3. **Mitigation:** Can simple adjustments — e.g. joint calibration of the DP clipping norm and the robust aggregation threshold — reduce the compounding effect without sacrificing formal guarantees?

1.4 Contributions

[Bullet list mirroring Section 6 of the Exposé: (i) systematic 8-configuration mapping on MNIST/CIFAR-10 with joint trilemma reporting; (ii) reproduction and cross-framework extension of BLADES baselines; (iii) initial test of joint clipping-norm calibration as a mitigation heuristic; (iv) open-source reproducible pipeline.]

1.5 Structure of the Report

The remainder of the report is organized as follows. Section 2 provides the theoretical background of all three trilemma axes and derives the joint lower bound. Section 3 defines the threat model and evaluation metrics. Section 4 describes the chosen mechanisms and their parametrization. Section 5 details the experimental setup and software stack. Section 6 presents and analyses the empirical results. Section 7 interprets findings, proposes mitigation, and tests the joint calibration heuristic. Section 8 addresses limitations and future work. Section 9 concludes.

2 Background

This chapter summarizes the theoretical background of FL, the trilemma axes and formalizes the Privacy-Robustness-Performance trilemma.

2.1 Federated Learning

Federated Learning is a machine learning framework comprising a central server and multiple distributed clients. Each client maintains a local dataset to compute updates for the global model. The central server orchestrates the training process by distributing the current global model, collecting client updates, and aggregating them to produce a new global model update. This update is then distributed to clients to initiate the next training round.

The most widely used aggregation method is FedAvg [1], which computes the global update as a weighted average of the client-local model updates.

2.2 Privacy: DP

This chapter introduces the concept of DP and the Byzantine Fault Model.

2.2.1 Definition and Composition

DP forms a good standard for privacy guarantees for algorithms used in data aggregation. Its degree of protection is defined by the privacy budget ε and the failure probability δ. We use the definition given by Abadi et al. [2].

Definition 1. *A randomized mechanism $M\colon D \to R$ with domain D and range R satisfies (ε, δ)-differential privacy if for any two adjacent inputs $d, d' \in D$ and for any subset of outputs $S \subseteq R$ it holds that*

$$\Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D') \in S] + \delta.$$

The privacy budget ε denotes the trade-off between utility and privacy, bounding the tolerated noise addition. Small values ($\varepsilon < 1$) provide the strongest privacy guarantees while larger values ($\varepsilon > 8$) allow for more accurate training.

The failure probability δ, introduced by Dwork et al. [6], denotes the probability that the plain ε-DP is broken and should be chosen smaller than the inverse size of the local dataset.

2.2.2 DP-SGD

DP-SGD is a modification to the standard Stochastic Gradient Descend (SGD) algorithm introduced by Abadi et al. [2] in 2016 to integrate DP into deep learning frameworks. It enhances SGD by clipping the computed gradient to a fixed norm and adding calibrated (Gaussian) noise before performing an update step. The algorithm pseudocode as proposed by Abadi et al. is shown in figure 1.

Algorithm 1 Differentially private SGD (Outline)

Input: Examples $\{x_1, \dots, x_N\}$, loss function $\mathcal{L}(\theta) = \frac{1}{N}\sum_i \mathcal{L}(\theta, x_i)$. Parameters: learning rate η_t, noise scale σ, group size L, gradient norm bound C.

Initialize θ_0 randomly

for $t \in [T]$ **do**

Take a random sample L_t with sampling probability L/N

Compute gradient

For each $i \in L_t$, compute $\mathbf{g}_t(x_i) \leftarrow \nabla_{\theta_t}\mathcal{L}(\theta_t, x_i)$

Clip gradient

$\bar{\mathbf{g}}_t(x_i) \leftarrow \mathbf{g}_t(x_i) / \max\left(1, \frac{\|\mathbf{g}_t(x_i)\|_2}{C}\right)$

Add noise

$\tilde{\mathbf{g}}_t \leftarrow \frac{1}{L}\left(\sum_i \bar{\mathbf{g}}_t(x_i) + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I})\right)$

Descent

$\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{\mathbf{g}}_t$

Output θ_T and compute the overall privacy cost (ε, δ) using a privacy accounting method.

Figure 1: DP-SGD pseudocode as published by Abadi et al. [2].

2.2.3 Central vs. Local DP in FL

[review]

Central DP is applied by the server by adding noise after the aggregation. This is used in settings where the aggregator is trusted, but updates need to be masked before sharing them with third parties.

In our case, the local updates are to be protected from inference attacks by the aggregator or attackers eavesdropping on the client-server-communication, requiring local DP. This is achieved by each client adding the noise locally before sharing the updates. The effective noise magnitude increases in this case because the masks on the individual updates add up, making local DP more costly in terms of accuracy.

2.3 Robustness: Byzantine-Tolerant Aggregation

2.3.1 Byzantine Fault Model

Byzantine Fault Tolerance is a property of a distributed system that allows for the system to stay operational while some components fail or act maliciously.

A η-fraction Byzantine adversary means that η% of the clients are acting maliciously. In the conducted experiments, malicious behavior means communicating manipulated gradients to poison the central model, e.g. insert a backdoor, degrade the general performance or enable label flipping attacks.

2.3.2 Krum

Krum [8], proposed by Blanchard et al. in 2017, enhances robustness to Byzantine adversaries by selecting updates closest to their XX neighbors, effectively focusing on the densest cluster of updates of size XX. This ensures the training process is provably Byzantine-tolerant but relies on the assumption that honest updates cluster together, which only holds when data distributions across clients are similar. Under data heterogeneity, this assumption breaks down, leading to performance degradation because honest updates may be discarded if they deviate significantly from the majority.

2.3.3 FLTrust

FLTrust [9] was introduced by Cao et al. in 2020 and mitigates malicious updates through trust scores. These are obtained by calculating the cosine similarity of the local update to an update anchor computed on a small root dataset held by the server. The method relies on the assumption that the server has access to a dataset representing the general data distribution across all clients.

2.4 Performance: Communication Efficiency

Another practical limitation especially in mobile applications is the available bandwidth to transmit local updates to the aggregator. We therefore include a method to communicate sparse instead of dense updates.

2.4.1 Top-k Gradient Sparsification

[review]

Top-k gradient sparsification [10] is a technique to reduce communication while uploading the gradients by only transmitting the k largest gradient components where k is a fixed fraction of the model dimension d.

This naturally degrades performance depending on the information distribution within the gradients. Models that allow for precise decisions along few axes are less affected than more complex ones that rely heavily on inter-dimensional relations.

2.4.2 Relation to Model Dimensionality

[The term $d/(\varepsilon^2 n)$ in the theoretical lower bound [3] shows efficiency is a fundamental third axis, not merely an engineering concern.]

2.5 The Privacy–Robustness–Performance Trilemma

2.5.1 Theoretical Lower Bound

[Allouah et al. [3]: The cross-term η/ε^2 shows multiplicative, not additive, interaction between privacy and robustness. SMEA (matches bound, exponential cost) vs. CAF (polynomial approximation).]

Allouah et al. [3] showed that any algorithm guaranteeing ε-local DP and robustness against an η-fraction Byzantine adversary incurs a mean-estimation error of at least $\Omega\big(\frac{\eta}{\varepsilon^2}+\frac{d}{\varepsilon^2 n}+\eta G^2\big)$.

The cross-term η/ε^2 shows that the cost for privacy and robustness interact multiplicative instead of additive.

2.5.2 Efficiency Axis Extension

[Position paper [4]: extends framework to efficiency; identifies theory–practice gap as open problem; calls for joint benchmarks.]

2.5.3 Empirical Evidence of Compounding

[Guerraoui et al. [11]: naive DP+Byzantine combination degrades performance beyond either alone. Lack of systematic 3-axis empirical mapping motivates this work.]

3 Threat Model and Metrics

This chapter maps out the assumed threat model and introduces the metrics used to evaluate the three axes.

3.1 Threat Model

We assume an honest-but-curious central server that adheres to protocol specifications but may attempt to infer private information from transmitted updates, motivating the use of differential privacy (DP) to safeguard client data. The adversary is characterized by a fraction η of malicious clients, capable of arbitrarily manipulating gradients (e.g., through label-flipping attacks, as formalized in Fang et al. [12]). These clients aim to corrupt the global model by injecting adversarial updates. Crucially, we assume no collusion between Byzantine clients and the server, restricting adversarial behavior to local manipulations of gradients. This assumption simplifies the analysis while maintaining practical relevance for decentralized learning systems.

3.2 Evaluation Metrics

As suggested by Allouah et al. [4], every configuration is evaluated simultaneously across all three trilemma axes:

Privacy. The privacy objective is quantified by the accumulated (ε,δ)-differential privacy (DP) budget after T rounds, tracked using the Rényi accountant [**?**]. Additionally, the vulnerability to membership inference attacks is assessed by measuring the success rate of such attacks against a shadow model baseline [**?**].

Robustness. Robustness is evaluated by the attack success rate (ASR) under label-flipping attacks and the accuracy gap Δ_{acc} relative to a clean FedAvg baseline. The accuracy gap captures untargeted performance degradation caused by adversarial clients, providing a measure of resilience to Byzantine behavior.

Performance. The performance metric includes test accuracy on held-out data, communication cost (bytes sent per client per round), and wall-clock training time. These metrics collectively assess the trade-off between model utility, resource efficiency, and computational feasibility in decentralized learning systems.

3.3 Trilemma Visualization

[Radar / simplex chart positioning each of the 8 configurations relative to the three axes, analogous to Figure 1 of [4]. Include actual figure once results are available.]

4 Methods

This section briefly describes and justifies the selected methods for each axis.

4.1 Baseline: FedAvg

[Standard weighted average aggregation [1]; serves as the reference point for all accuracy gaps; no privacy, robustness, or compression applied.]

FedAvg, introduced by McMahan et al. in 2017 [1], is a widely used aggregation strategy in FL designed to handle non-IID data efficiently. It computes a weighted average of local model updates, with weights determined by the size of each client's local dataset. The paper establishes baselines on standard datasets such as MNIST and CIFAR-10, making FedAvg a foundational reference for evaluating FL methods. Its simplicity and effectiveness in early FL research have cemented its role as a standard baseline for comparative studies.

4.2 Privacy Mechanism: DP-SGD

[Per-sample clipping norm C; noise multiplier σ; privacy budget $\varepsilon \in \{1,5,10\}$, $\delta = 10^{-5}$; Rényi accounting via Opacus [7].]

4.3 Robustness Mechanism: FLTrust

[Server root dataset construction; trust score computation; cosine similarity clipping; parameter: root dataset size; implementation via BLADES [13].]

The root dataset for the central server was obtained by randomly sampling XX datapoints from WHERE?

The trust score for weighting the local contributions is calculated as the cosine similarity to the update vector that was computed from the root data set as depicted in figure 2.

4.4 Performance Mechanism: Top-k Sparsification

[Fraction $k \in \{1\%, 10\%\}$ of d transmitted per round; error feedback accumulator; custom Flower strategy wrapper; interaction with gradient clipping discussed in Section 4.5.]

4.5 Mechanism Interaction Effects

[Theoretical analysis of how clipping (DP) and cosine-similarity clipping (FLTrust) may amplify each other; how sparsification changes the effective gradient distribution seen by the aggregator; hypothesis: super-additive cost for the full combination.]

5 Experimental Setup

5.1 Experimental Grid

The experimental grid is described in table 1.

Identifier	DP-SGD	FLTrust	Top-k
Base			
DP	x		
ROB		x	
COM			x
DP+ROB	x	x	
DP+COM	x		x
ROB+COM		x	x
ALL	x	x	x

Table 1: Identifiers and applied methods for the different experimental setups.

5.2 Datasets

[MNIST: simple baseline, reproducibility check against BLADES published numbers; image size, number of classes, train/test split. CIFAR-10: scalability test; used in the position paper's empirical evidence table [4]; image size, number of classes, train/test split. Data partitioning strategy across clients (IID / Dirichlet non-IID).]

5.3 Fixed Hyperparameters

[Justify each choice with respect to BLADES baselines.]

Number of clients n; Byzantine fraction η; number of communication rounds T; local epochs; local batch size; learning rate; model architecture (CNN for MNIST, ResNet-*[X]* for CIFAR-10).

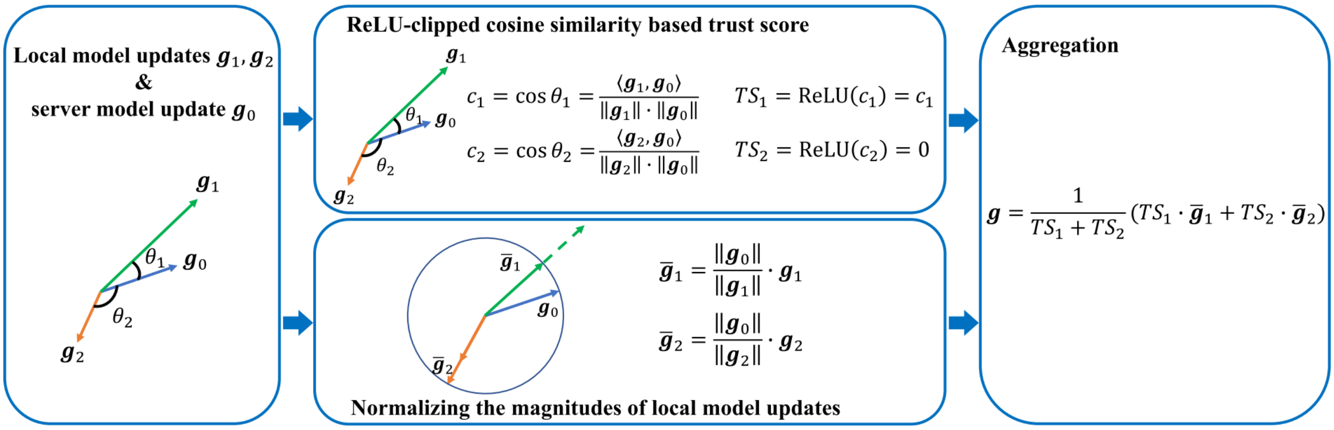

Figure 2: Global aggregation rule for FLTrust as introduced by Cao et al. [9].

5.4 Software Stack

- **Flower** (flwr) [14] — FL orchestration, client/server simulation, strategy interface.
- **Opacus** [7] — DP-SGD, Rényi DP accounting, gradient clipping.
- **FLTrust** [9] — FLTrust defense implementation.
- **BLADES** [13] — Byzantine attack implementations; PyTorch/Ray backend compatible with Flower.
- **Custom** — Top-k gradient sparsification wrapper.

[Software versions; hardware (GPU/CPU); random seed management; number of independent runs per configuration.]

5.5 Reproducibility

[Pointer to open-source repository; instructions to replicate all 8 configurations; fixed random seeds; containerization (Docker/Conda environment file).]

6 Results

6.1 Baseline Reproduction (RQ1 – Sanity Check)

[FedAvg and FLTrust baselines on MNIST vs. published BLADES numbers; table: our accuracy / ASR vs. reported values; discussion of any deviations.]

6.2 Isolation: Individual Mechanism Costs (RQ1)

[One subsection or figure per mechanism; accuracy vs. ε curve for DP-SGD; ASR vs. η for FLTrust; accuracy / bytes trade-off for Top-k at $k \in \{1\%, 10\%\}$; results on both MNIST and CIFAR-10.]

6.2.1 DP-SGD Cost

[Accuracy curves for $\varepsilon \in \{1, 5, 10\}$; privacy budget consumed.]

6.2.2 FLTrust Cost

[Accuracy gap and ASR for varying Byzantine fractions.]

6.2.3 Top-k Sparsification Cost

[Accuracy, bytes per round, and wall-clock time for $k \in \{1\%, 10\%\}$.]

6.3 Combination: Pairwise and Full Interactions (RQ2)

[Main results table: 8 configurations $\times$ all metrics; additive vs. super-additive cost analysis; focus on DP+FLTrust interaction; radar / simplex visualization (Figure [X]).]

6.3.1 DP + Robustness

[Effect on accuracy and ASR; comparison to individual costs.]

6.3.2 DP + Compression

[Effect on accuracy and communication cost; interaction with noise.]

6.3.3 Robustness + Compression

[Effect on ASR and bytes per round.]

6.3.4 Full Combination (DP + Robustness + Compression)

[Worst-case configuration; super-additive compounding quantified.]

6.4 MNIST vs. CIFAR-10 Comparison

[How results scale with dataset complexity; does the compounding pattern hold across both?]

7 Discussion

7.1 Interpretation of Compounding Effects

[Relate empirical findings to the theoretical cross-term η/ε^2; characterize compounding as additive / sub-additive / super-additive per configuration pair; intuition for why DP clipping and FLTrust cosine clipping interact destructively.]

7.2 Mitigation: Joint Clipping-Norm Calibration (RQ3)

[Heuristic: jointly set DP clipping norm C and FLTrust trust threshold to minimize redundant gradient suppression; description of the calibration procedure; results vs. uncalibrated worst-case combination; formal guarantee preservation argument.]

7.3 Practical Recommendations

[When to prioritize which axis; safe operating region in the trilemma space; guidance on (ε, k) parameter selection given a robustness requirement.]

8 Limitations and Future Work

8.1 Limitations

[Single attack type (label-flipping); single aggregation rule (FLTrust); membership inference approximation (shadow model only); limited ε grid; MNIST/CIFAR-10 may not represent medical/financial FL.]

The experiments in this project were conducted under several simplifications to provide a lightweight anchor for future research. The datasets and models selected offer good comparability with existing work but may not fully represent the data analyzed in practical FL applications (e.g., financial or medical domains) or complex architectures (e.g., transformer-like models).

Furthermore, we evaluated only a single method and at most one attack type per axis. The baseline for membership inference attack success rate was approximated using a shadow model, and the privacy budget was constrained to a limited ε-grid to balance computational feasibility with analytical rigor. Additionally, the privacy budget was spent uniformly over time, but recent research by Kiani et al. [15] suggests that a time-adaptive approach yields better privacy-utility-tradeoffs.

8.2 Future Work

[Extend to Krum, SMEA, CAF aggregators; stronger attacks (model poisoning [12]); LoRA+DP reparametrization [16] for the efficiency axis; theoretical tightness of joint calibration heuristic.]

Future work should aim to expand the experimental setup to encompass a broader range of methods (e.g., Krum, SMEA, CAF for aggregation) and evaluate stronger adversarial scenarios, such as model poisoning attacks described by Fang et al.[12]. Additionally, exploring reparametrization techniques such as the Low-Rank-adaptation method proposed by Liu et al.[16] could provide insights into balancing efficiency and privacy in FL systems.

Other promising directions include varying the Byzantine fraction, testing alternative threat models (e.g., malicious servers, collusion between malicious clients and servers), or introducing data heterogeneity through artificially distributed datasets. These extensions would better reflect real-world complexities and enable more robust comparisons across methodologies..

9 Conclusion

[Restate research questions and answers; key quantitative finding (e.g. full combination degrades accuracy by X% beyond sum of individual costs); mitigation heuristic reduces compounding by Y% without formal guarantee loss; open-source pipeline as contribution to the community.]

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